Simplified Cost Function for Logistic Regression

To make implementation easier, the two-part logistic loss function can be combined into a single, equivalent equation.

Simplified Loss Function

Recall the original loss function was defined in two parts, one for when $y = 1$ and another for when $y = 0$. Because y in a binary classification problem can *only* be 0 or 1, we can combine these into one expression.

- **The simplified loss function** for a single training example is:
  $L(f(\vec{x}), y) = -y \log\big(f(\vec{x})\big) - (1 - y) \log\big(1 - f(\vec{x})\big)$
- **How it Works:**
  - **Case 1: True label is 1 ($y = 1$)**
    - The second term, $(1 - y) \log(\ldots)$, becomes $0 \cdot \log(\ldots) = 0$.
    - The equation simplifies to $L = -1 \cdot \log\big(f(\vec{x})\big)$, which is the original loss for $y = 1$.
  - **Case 2: True label is 0 ($y = 0$)**
    - The first term, $-y \log(\ldots)$, becomes $0 \cdot \log(\ldots) = 0$.
    - The equation simplifies to $L = -(1 - 0) \log\big(1 - f(\vec{x})\big)$, which is the original loss for $y = 0$.

This single-line formula is mathematically identical to the previous piecewise definition.

Full Cost Function using Simplified Loss

The overall cost function, $J(\vec{w}, b)$, is the average of this simplified loss function over all m training examples.

- **Final Cost Function for Logistic Regression:**
  $J(\vec{w}, b) = -\frac{1}{m} \sum_{i=1}^{m} \left[y^{(i)} \log\big(f_{\vec{w},b}(\vec{x}^{(i)})\big) + (1 - y^{(i)}) \log\big(1 - f_{\vec{w},b}(\vec{x}^{(i)})\big) \right]$
- **Rationale for this Function:**
  - This cost function is derived from a statistical principle known as **Maximum Likelihood Estimation**.
  - Most importantly, it has the desirable property of being **convex**, which ensures that gradient descent can reliably find the global minimum.